

Nimisha Karnatak

Ph.D. in Computer Science at University of Oxford

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EDUCATION

University of Oxford, United Kingdom

Oct 2023 - Present

Doctor of Philosophy in Computer Science

- Supervisors: [Professor Sir Nigel Shadbolt](#) and [Professor Max Van Kleek](#)
- Publications: CHI 2026 [1], CHI 2025 (W) [2], AAAI 2024 (W) [4]
- Distinction awarded for Assessed Work at Transfer of Status

International Institute of Information Technology, Bangalore

July 2019 - June 2021

Master of Science in Digital Society, Specialization: Data Intensive Digital Design

CGPA: 3.72/4.0

- Dean's Merit List of IIIT-B (For outstanding academic performance)

G.B. Pant University of Agriculture & Technology

Sept 2014 - June 2018

Bachelor of Technology in Industrial and Production Engineering

CGPA: 8.093/10.0

- Vice Chancellor's Silver Medalist of the batch 2014 - 18 (Awarded to the highest CGPA holder)
- Toppers merit scholarship throughout the B.Tech. programme

WORK EXPERIENCE

The World Bank

Washington, D.C., United States (*Remote*)

Research Intern

May 2025 – April 2026

Advisors: [Dr. Renos Vakis](#), [Zeina Afif](#)

Publication: ACM CHI 2026 [1]

Google DeepMind London (AIUX)

London, United Kingdom

Student Researcher

April 2024 - Oct 2024

Advisor: [Kristen Olson](#)

Publication: Deepmind Technical Preprint [3] and ACM CHI Workshop [2]

Microsoft Research India

Bangalore, India

Research Fellow (Full-Time Position)

July 2021 - May 2023

Collaborators: [Dr. Bill Thies](#), [Dr. Mohit Jain](#), [Dr. Indrani Medhi Thies](#)

Publications: ACM TOCHI [5] and ACM CSCW [6]

PUBLICATIONS

- Learning from AVA: Early Lessons from a Curated and Trustworthy Generative AI for Policy and Development Research**
Nimisha Karnatak, Mohamad Chatila, Daniel Pinzon, Reza Yazdanfar, Michelle Dugas, Renos Vakis.
Accepted at ([ACM CHI 2026](#)) [Paper] [Video Submission] and [AVA (AI) Official Website – The World Bank]
- Expanding the Generative AI Design Space through Structured Prompting and Multimodal Interfaces.**
Nimisha Karnatak, Adrien F. Baranes, Rob Marchant, Huinan Zeng, Triona Butler, Kristen Olson.
Accepted at the ACM SIGCHI Conference on Human Factors in Computing Systems, Workshop on Designing and Developing User Interfaces with AI ([ACM CHI 2025](#)) [PDF]
- ACAI for SBOs: AI Co-Creation for Advertising and Inspiration for Small Business Owners**
Nimisha Karnatak, Adrien F. Baranes, Rob Marchant, Triona Butler, Kristen Olson.
([Google Deepmind](#)) [PDF]
- Generative AI as Collaborative Technology for ASHA Workers in Marginalized Communities**
Nimisha Karnatak, Max Van Kleek, Nigel Shadbolt.
Accepted at the Collaborative AI and Modeling of Humans Bridge at the Association for the Advancement of Artificial Intelligence ([AAAI 2024](#)) [PDF]
- “Is it even giving the correct reading or not?”: How Trust and Relationships Mediate Blood Pressure Management in India**
Nimisha Karnatak, Brooke Loughrin, Tiffany Kuo, Odeline Silvernail, Indrani Thies, Bill Thies, and Mohit Jain
Accepted paper at the Transactions on Human-Computer Interactions ([ACM TOCHI](#)).
Note: Presented at the ACM conference on Human Factors in Computing Systems [ACM CHI 2024](#) [PDF]
- Understanding the Technology-Mediated Home Phlebotomy Ecosystem in India**
Nimisha Karnatak*, Meghna Gupta* and Mohit Jain (* denotes equal authorship)
Accepted paper at the ACM Conference On Computer-Supported Cooperative Work And Social Computing ([CSCW 2024](#)) [PDF]

INVITED TALKS

Google DeepMind, London, August 2024: Bridging the Expertise Gap: Understanding the Dynamics of Co-Creation between LLM-powered Tools and Novice Designers

UXR Health Collectives, Microsoft Research Redmond, May 2022: “Is it even giving the correct reading or not?”: How Trust and Relationships Mediate Blood Pressure Management in India

Technology and Empowerment Group, Microsoft Research India, April 2022: Understanding the Ecosystem of Blood Pressure Management in Low Resource Communities of India

Oxford Women in Computer Science (OxWoCS), May 2025: Invited panelist at “Keep It Real”, the flagship event for women in STEM hosted by OxWoCS.

ACHIEVEMENTS

- Research Delegate, Citadel and Citadel Securities PhD Summit: **Selected among 25 globally distinguished PhD and postdoctoral researchers** to attend an invite-only summit on academic research applications in finance.
- Nominated by the University of Oxford for the **Google PhD Fellowship** through a competitive, university-wide selection process. ([Link](#))
- Nominated by the University of Oxford for the **Apple Scholars in AI/ML PhD Fellowship** through a competitive, university-wide selection process. ([Link](#))
- Awarded \$1,500 scholarship to present my work titled “Generative AI as Collaborative Technology for ASHA Workers in Marginalized Communities.” at the 38th AAAI Conference on Artificial Intelligence in person
- Selected for the SIGCHI Emerging Scholars at CSCW 2023 and awarded \$1,000 in financial support.
- Amongst the top 65 students out of over 15000 applicants across India to get selected as a **SPARK summer Intern for 2020** at the Indian Institute of Technology, Roorkee (**IIT-R**)
- Won the ‘**Best Presentation Award**’ for my internship work at the SPARK Internship Program of the Indian Institute Of Technology, Roorkee (**IIT-R**) [Certificate Of Appreciation by IIT-R and SPARK](#)

FUNDED ACADEMIC RESEARCH

Indian Institute of Technology Roorkee <i>User Experience Research Intern</i>	Roorkee, India June 2020 - July 2020
Project: Evaluation of urban space from the perception of the elderly to build an inclusive neighborhood	
Advisor: Prof. Saptarshi Kolay (IIT-Roorkee)	
International Labour Organisation (ILO) <i>Qualitative Research Assistant</i>	Geneva, Switzerland Sep 2019 - June 2020
Project: Exploring the experiences of women working in the AI domain in India and understanding the ethical aspects of AI from the perspective of women.	
Advisors: Dr. Uma Rani (Senior Economist, ILO), Prof. Preeti Mudliar (IIIT-B)	
Digital Identity Research Initiative at Indian School Of Business and IIIT-B <i>Research Intern</i>	Bangalore, India Jan 2020 - June 2020
Project: Exploring the implication of Aadhar failure and consequential exclusion.	
Advisor: Prof. Bidisha Chaudhuri (IIIT-B)	

ACADEMIC & COMMUNITY SERVICES

Graduate Teaching Assistant <i>Artificial Intelligence (AI), University of Oxford</i>	Jan 2026 - March 2026
Mentor <i>Supervising 6 undergraduate Computer Science students, University of Oxford</i>	Feb 2025 - May 2025
Graduate Teaching Assistant <i>Interaction Design (IDE), University of Oxford</i>	November 2024
External Reviewer <i>ACM CHI Conference on Human Factors in Computing Systems (ACM CHI)</i>	2024,2025,2026

ACADEMIC PROJECTS

Few-Shot Classification of Medical Data: Exploring Self-Supervised Learning Approaches for Classifying Low Resource Medical Data: A Comparative Investigation With MedMNIST

- Used a hybrid approach leveraging self-supervised learning and active learning to perform few-shot image classification of medical data; Technology Stack: Python, PyTorch, Numpy, Matplotlib

Brain MRI Segmentation: Developed a deep-learning network for semantic segmentation of brain MRI scans.

- Trained a UNet-based architecture using Dice loss as objective function to perform brain MRI segmentation; Technology Stack: Python, PyTorch, Numpy, Matplotlib

User Research to Design Accessible Mobile Payment Apps for the Visually Challenged:: Developed low-fidelity prototype (Braille inscribed on cardboard simulating mobile phone screens) and high-fidelity prototype (by deploying Adobe XD) using design concepts from user research for usability testing

- Research methodology: qualitative interviews of 30-40 minutes with 11 visually challenged participants
- Usability Testing methods: Think out loud and Participant observation followed by an interview

Developed high-fidelity virtual classroom prototype using Adobe XD: Analyzed the impact of emergency remote learning on IIIT-B students during COVID-19 through quantitative and qualitative research.

Spatial Data Analysis of Bangalore's Land Use and Cover: Conducted a Time Series Analysis on Bangalore's Land Use and Land Cover using GRASS GIS software for the years 2003, 2010, 2015, and 2020.

SKILLS / TECHNOLOGIES

Human Computer Interaction: User Study, Focus Groups, Prototyping, Wireframing, Adobe XD, Usability Testing and Evaluation, Design Thinking, Accessible and Inclusive design, Personas, Scenarios, Storyboards, Diary study, Survey Design and Sampling, Stakeholder/user interviews, Figma

Research and Data Analysis Skills: RCTs and Causal inference, Qualitative Research & Quantitative Research, Data Collection & Analysis, Participant Observation, Ethnographic Research, Randomized Controlled Trial, Microsoft Excel, Advanced Statistics

Technical Skills: Python (NumPy), PyTorch, LLM Evaluation, LLM-as-Judge Evaluation, RAG Evaluation, Prompting, Safety Evaluation Rubrics